$\sum \left(\frac{n+1}{2n+1}\right)^n$, $\sqrt[n]{\left(\frac{n+1}{2n+1}\right)^n} = \frac{n+1}{2n+1} = \frac{1+1/n}{2-1/n} \to 1/2 < 1$

$\hookrightarrow$ CV.

$\sum (-1)^n a_n$

$\sum_{n\geq 0} (-1)^n$, $\sum_{n=0}^{N} (-1)^n = \begin{cases} 0 & \text{si } N \text{ es impar} \\ 1 & \text{si } N \text{ es par} \end{cases}$

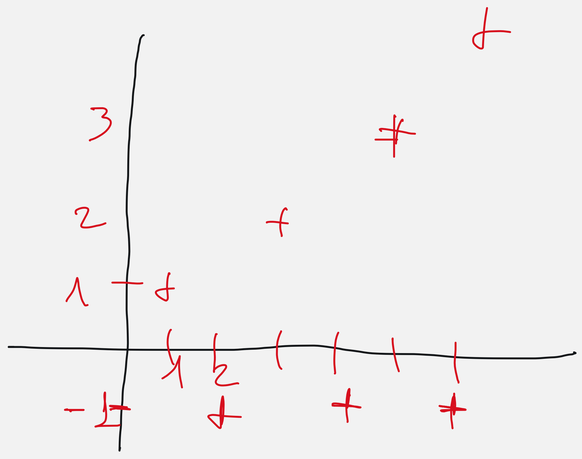
$\sum 1 = +\infty$

$(-1)^0 = 1$

$(-1)^0 + -1 = 0$

$+1-1+1 = 1$

$(-1)^n \underbrace{a_n}_{\geq 0}$



$a_n = \begin{cases} -1 & \text{si } n \text{ es par} \\ +\frac{n-1}{2} & n \text{ impar} \end{cases}$

$\sum_{n=0}^{N} n = \frac{n(n+1)}{2}$

$\underbrace{\sum_0^N a_n}_{DV} = (-1)\frac{N}{2} + \underbrace{\frac{1}{2}\sum^{N} \frac{n-1}{2}}_{\sim C \cdot N^2,\ C>0}$

$(-1)^n b_n$

$b_n = \begin{cases} \frac{n-1}{2} & \text{si } n \text{ es impar} \\ +1 & \text{si } n \text{ es par} \end{cases}$

$\approx C \cdot N^2 - C'N \to +\infty$

$\sum |a_n|$ CV. (?)

$o(N)$ $\bullet(N^2)$ $o(\exp(N) \times 1/N)$

$\sum_{n\geq 1} \frac{n}{(n+1)\log(n+1)} \simeq \sum \left(\log(n+1)\right)^{-1}$

$\frac{n}{(n+1)} \times \frac{1}{\log(n+1)}$

$\to 1$ $\to 0$

$\sum \frac{1}{a_n}$, $\sum_{n\geq 1} \frac{1}{n}$ DV.

$\log(n) \leq n \Rightarrow \frac{1}{\log(n)} \geq \frac{1}{n}$

$\Rightarrow \sum^{N} 1/\log n \geq \sum^{N} 1/n \to +\infty$